FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FIITJEE

JEE(Main)-2015

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CRT –II
(Main)

S. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	A	D
2.	B	B	B
3.	B	D	C
4.	D	B	A
5.	C	C	C
6.	B	A	D
7.	A	C	B
8.	B	B	A
9.	D	B	A
10.	D	A	B
11.	B	A	C
12.	C	C	B
13.	C	A	A
14.	B	D	B
15.	B	B	C
16.	C	C	B
17.	A	A	A
18.	C	B	C
19.	B	C	B
20.	B	B	C
21.	D	D	D
22.	C	A	D
23.	A	D	B
24.	D	A	D
25.	D	D	C
26.	B	B	B
27.	A	B	B
28.	D	C	A
29.	D	D	B
30.	B	D	D

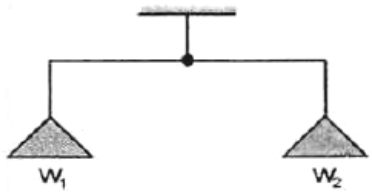
Physics

PART – I

SECTION – A

1. Since the compartment (including passengers) is stationary, so the combined centre of mass of compartment and passengers is fixed i.e. C_2 is fixed. When the passengers move here and there in the compartment then in an attempt to keep C_2 fixed C_1 has to move.

2. Since the balance has equal arms and it is a false balance, so the pans must be having different weights. Let w_1 and w_2 be the weights of left pan and right pan respectively. Let w be the true weight of object. When placed in left pan it weights x i.e. a total weight of $w + w_1$ on left balances a total weight of $w_2 + x$ on the right pan.



$\Rightarrow w + w_1 = w + x$ .....(1)

Similarly it weighs y when placed in right pan i.e.

$y + w_1 = w + w_2$ .....(2)

From (1)

$w_1 - w_2 = x - w$ .....(3)

From (2)

$w_1 - w_2 = w - y$ .....(4)

$\Rightarrow x - w = w - y$

$\Rightarrow w = \frac{x+y}{2}$

3. The center gravity of liquid in the first column is $\frac{h_1}{2}$ and that in the second column is $\frac{h_2}{2}$.

Let litial Potential Energy be U_i

$\Rightarrow U_i = (Ah_1\rho)g\left(\frac{h_1}{2}\right) + (Ah_2\rho)g\left(\frac{h_2}{2}\right)$

$\Rightarrow U_i = \frac{1}{2}A\rho g(h_1^2 + h_2^2)$

When the water column in the two equavalises then equivalent height is $\frac{h_1+h_2}{2}$ and the center of gravity in both the columns is $\frac{h_1+h_2}{4}$. If U_f be the final potential energy in both the columns then

$U_f = 2\left[A\left(\frac{h_1+h_2}{2}\right)\rho\left(\frac{h_1+h_2}{4}\right)g\right]$

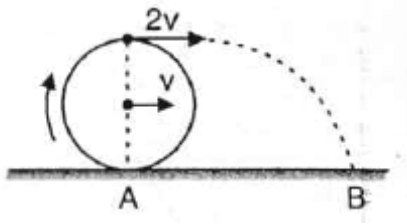
$\Rightarrow U_f = \frac{1}{4}A\rho g(h_1+h_2)^2$

Since work done is equal to the decrease in potential energy,

So, $w = U_i - U_f$

$\Rightarrow w = A\rho g\left(\frac{h_1-h_2}{2}\right)^2$

5. From the concepts of projectile motion,



$2r = \frac{1}{2}gt^2$

Since $x = AB = (2v)t$

$\Rightarrow t = \sqrt{\frac{4r}{g}}$

$\Rightarrow AB = 4v\sqrt{\frac{r}{g}}$

6. $y = ax^2$

$\Rightarrow \frac{dy}{dt} = \dot{y} = (2ax)\dot{x} = 2acx \qquad \left\{\because \frac{dx}{dt} = \dot{x} = c\right\}$

Further $a = \frac{d^2y}{dt^2} = \ddot{y} = 2ac\dot{x} = 2ac^2$

7. $(0)A = 4v + (A - 4)V$

$\Rightarrow V = -\frac{4v}{A-4}$

Negative sign indicates the recoil speed.

8. Let the extension produced in the spring due to rotation be x.
Then, for the body to be in equilibrium

$F_{spring} = F_{centrifugal}$

$\Rightarrow kx = m(\ell + x)\omega^2$

$\Rightarrow x = \frac{m\ell\omega^2}{k - m\omega^2}$

10. $\vec{E} = \vec{\nabla}V$

$\Rightarrow \vec{E} = -\left(\hat{i}\frac{\partial U}{\partial x} + \hat{j}\frac{\partial U}{\partial y} + \hat{k}\frac{\partial U}{\partial z}\right)$

$\frac{\partial U}{\partial x} = 6 - 8y^2 - 8x$

$\frac{\partial U}{\partial y} = -16xy - 8 + 6z$

$\frac{\partial U}{\partial z} = 6y$

At (0, 0, 0)

$\frac{\partial U}{\partial x} = 6\,,\ \frac{\partial U}{\partial y} = -8\,,\ \frac{\partial U}{\partial z} = 0$

$\Rightarrow |\vec{E}|_{(0,0,0)} = -6\hat{i} + 8\hat{j}$

$\Rightarrow |\vec{E}| = 10NC^{-1}$

$\Rightarrow |\vec{F}| = q|\vec{E}|$

$\Rightarrow |\vec{F}| = 20\,N$

11. Since $P_1V = n_1RT_1$
and $P_2V = n_2RT_2$
After joining the vessels we have
$P(2V) = (n_1 + n_2)RT$

$$\Rightarrow P(2V) = \left(\frac{P_1V}{RT_1} + \frac{P_2V}{RT_2}\right)RT$$

$$\Rightarrow P = \frac{1}{2}\left(\frac{P_1}{T_1} + \frac{P_2}{T_2}\right)T$$

12. $$\frac{0.693}{\lambda_X} = \frac{1}{\lambda_Y}$$

$\Rightarrow \lambda_Y > \lambda_X$

Since, $\left(-\frac{dN}{dt}\right)_X = \lambda_X N$

And $\left(-\frac{dN}{dt}\right)_Y = \lambda_Y N$

$$\Rightarrow \left(-\frac{dN}{dt}\right)_Y > \left(-\frac{dN}{dt}\right)_X$$

Decay rate of Y > Decay rate of X.

15. $$v_d = \frac{1}{neA}$$

16. Soft iron has highest susceptibility.

17. $\lambda_m T$ = constant

$$\Rightarrow T = \frac{4}{3}T_0$$

Since, $P \propto T^4$ (from Stefan's Law)

$$\Rightarrow P_{new} = \frac{256}{81}P$$

18. $$\tan\theta_1 = \frac{\tan\theta}{\cos\alpha}$$

and $$\tan\theta_2 = \frac{\tan\theta}{\cos(90^\circ - \alpha)} = \frac{\tan\theta}{\sin\alpha}$$

$$\Rightarrow \cos\alpha = \frac{\tan\theta}{\tan\theta_1} \quad \text{.....(1)}$$

and $$\sin\alpha = \frac{\tan\theta}{\tan\theta_2} \text{.....(2)}$$

Dividing (2) by (1), we have

$$\tan\alpha = \frac{\tan\theta_1}{\tan\theta_2}.$$

19. A and C are statements of Lenz's Law. B is the correct option.

20. $I_v \sin\phi$ component of current gives no power consumption and hence is called wattless current.

21. $v^2 = \omega^2(a^2 - x^2)$

$\Rightarrow v_1^2 = \omega^2(a^2 - y_1^2)$ .....(1)

$\Rightarrow v_2^2 = \omega^2(a^2 - y_2^2)$ .....(2)

From (1) and (2), we get

$$T = 2\pi\sqrt{\frac{y_1^2 - y_2^2}{v_2^2 - v_1^2}}$$

22. $$n = 2\left(\frac{v}{2\ell}\right)$$

$$n' = 3\left(\frac{v}{4\ell}\right) = \frac{3n}{4}$$

23. Resolving power of telescope = $\frac{a}{1.22\lambda}$

where a is the diameter of objective lens and λ is the wavelength of light used. On increasing a, more light is collected by objective lens and so, the image formed is more bright. Thus, resolving power of telescope increases.

24. $$\beta = \frac{\lambda D}{d}$$

So, angular with, 2θ is

$$2\theta = \frac{2\beta}{D} = \frac{2\lambda}{d}$$

$$\theta = \frac{6328 \times 10^{-10}}{0.2 \times 10^{-3}}$$

$\Rightarrow \theta = 3164 \times 10^{-6}$ radian

$\Rightarrow \theta = 3164 \times 10^{-6} \times \frac{180}{\pi}$ degree

$\Rightarrow \theta = 0.18$ degree

$\Rightarrow 2\theta = 0.36$ degree.

25. If I is the intensity of the incident unpolarised light, the intensity transmitted by the first is $\frac{I}{2}$. This is the intensity of incident light on the second polaroid. Intensity transmitted by the second polaroid is $\left(\frac{I}{2}\right)\cos^2\theta$, where θ is the angle between the axes. Here $\sin\theta = \frac{3}{5}$, $\cos\theta$ is therefore $\frac{4}{5}$.

$$\frac{I}{2}\cos^2\theta = \frac{I}{2} \times \left(\frac{4}{5}\right)^2 = \frac{8}{25}I$$

$\frac{8}{25}$ is the required ratio.

26. 50 VSD = 49 MSD

1 VSD = $\frac{49}{50}$MSD

$\Rightarrow$ VC = 1 MSD – 1 VSD

$\Rightarrow VC = 1\text{MSD} - \frac{49}{50}\text{MSD}$

$\Rightarrow VC = \frac{1}{50}MSD$

$\Rightarrow 1\ MSD = 50\ (VC) = 50\ (0.001\ cm)$

$\Rightarrow 1\ MSD = 0.05\ cm = 0.5\ mm$

$\Rightarrow 1\ MSD = 0.5\ mm$

27. $I_C = I_E - I_B = 4\ mA - 40\ \mu A = (4 - 0.04) = 3.96\ mA$

$\alpha = \frac{I_C}{I_E} = \frac{3.96}{4} = 0.99$

28. Electric energy density

$u_0 = \frac{1}{2}\varepsilon_0 E_{rms}^2$

Since, $E_{rms} = \frac{E_0}{\sqrt{2}}$

$\Rightarrow u_0 = \frac{1}{4}\varepsilon_0 E_0^2$

29. By the Law of conservation of energy

$(U + K)_{surface} = (U + K)_\infty$

$\Rightarrow \frac{-GMm}{R} + \frac{1}{2}m(3v_1)^2 = 0 + \frac{1}{2}mv^2$

$\Rightarrow -\frac{GM}{R} + \frac{9v_e^2}{2} = \frac{1}{2}v^2$

Since, $v_e^2 = \frac{2GM}{R}$

$\Rightarrow -\frac{v_e^2}{2} + \frac{9v_e^2}{2} = \frac{1}{2}v^2$

$\Rightarrow v^2 = 8v_e^2$

$\Rightarrow v = 2\sqrt{2}v_e$

30. Let h be the height of the water level at any instant. Then the rate of decrease of water level is $-\frac{dh}{dt}$. Therefore

$-A\frac{dh}{dt} = av = a\sqrt{2gh}$

$\Rightarrow -\frac{dh}{dt} = \frac{a}{A}\sqrt{2gh}$

$\Rightarrow -\int_{H_1}^{H_2}\frac{dh}{h^{1/2}} = \left(\frac{a}{A}\sqrt{2g}\right)\int_0^1 dt$

$\Rightarrow -2\left[h^{1/2}\right]_{H_1}^{H_2} = \left(\frac{a}{A}\sqrt{2g}\right)t$

$\Rightarrow 2\left[\sqrt{H_1} - \sqrt{H_2}\right] = \left(\frac{a}{A}\sqrt{2g}\right)t$

$\Rightarrow t = \frac{A}{a}\sqrt{\frac{2}{g}}\left[\sqrt{H_1} - \sqrt{H_2}\right]$

Chemistry

PART – II

SECTION – A

1. Salicylic acid contains – OH group and – COOH group. Therefore $FeCl_3$ and $NaHCO_3$ tests are used.

2. ZnO gives yellow colour on heating and white on cooling.

3. $PbCl_2 + K_2CrO_4 \longrightarrow \underset{\text{yellow ppt.}}{PbCrO_4} + 2KCl$

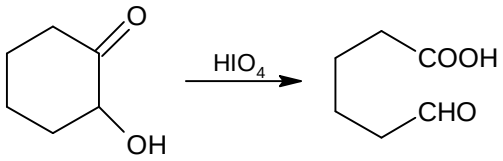
4. Dissociation of $H_2O(H_2O \rightleftharpoons H^+ + \bar{O}H)$ is opposite to neutralization. Hence, enthalpy of dissociation is equal to enthalpy of neutralization but has opposite sign.

5. For preparation of iodoform reagent which can supply I^+ is require i.e. KOI or KOH/I_2.

6. I^- ion is more easily oxidized than Br^- ion and hence violet colour due to I_2 appears first.

7. $AuCl_3$ is reduced by $SnCl_2$ to form colloidal sol of gold.
There is no peptisation involved.

8. In accordance with Huckel rule, carbocation (III) is aromatic and hence most stable. Carbocation (IV) is stabilized by resonance and hence more stable than carbocation (I and II). In carbocation (II), +ve charge is present on a more electronegative sp-hybridized carbon while in carbocation (I), it is present on a less electronegative sp^2-hybridised carbon and hence carbocation (I) is more stable than carbocation (II). Thus, the overall stability increases in the order : II < I < IV < III.

9. 

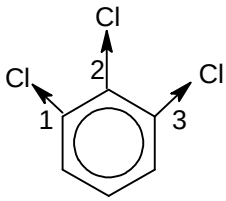
10. SeO_2 oxidizes – CH_2 – next to carbonyl group to another $\overset{O}{\overset{\|}{C}}$ group.

$$CH_3-\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow{SeO_2} \underset{x}{CH_3-\overset{O}{\overset{\|}{C}}-CHO} + Se + H_2O$$

11. 1 Mole of $C_{60}H_{122} = 60 \times 12\ g + 122 \times 1\ g = 842\ g$
$= 6.022 \times 10^{23}$ molecules
Weight of 1 molecule = $(842/ 6.022 \times 10^{23})$ gram.

12. $2C_6H_5COOH \rightleftharpoons (C_6H_5COOH)_2$
Before asso. 1 mol
After asso. 1 – x x/2
Total = 1 – x + x/2 = 1 – x/2

$$i = \frac{1-\frac{x}{2}}{1} = 1-\frac{x}{2}$$

13. 

, resultant bond moment of C_1 – Cl and C_3 – Cl bonds adds up to the bond moment of C_2 – Cl. Hence, it has maximum value.

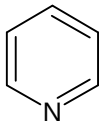
14. NO_2^+ has no unshared electron. It has only bond pairs in two directions. Hence, shape is linear $O{=}\overset{+}{N}{=}O$ with bond angle = 180º.

15. For reaction (i), $K_p = P_{CO_2} = 8 \times 10^{-2}$ (given)

For reaction (ii), $K_p = \dfrac{P_{CO}^2}{P_{CO}} = 2$ (given)

$\therefore \quad 2 = \dfrac{P_{CO}^2}{8 \times 10^{-2}}$ or $P_{CO} = 0.4$

16. Boron is diagonally related to Si.

17. In 

, N is linked to two c-atoms and has one lone pair. Hence, electron geometry is triangular planar and hybridization is Sp^2.

18. $E_2 = -\dfrac{k}{2^2} = -328 \text{ kJ mol}^{-1}$

$K = 1312 \text{ kJ mol}^{-1}$

$E_4 = -\dfrac{k}{4^2}$

$= -\dfrac{1312}{16} = -82 \text{ kJ mol}^{-1}$

19. Among isoelectronic ions, size increases the –ve charge increases.
i.e. $F^- < O^{2-} < N^{-3}$

20. Greater the boiling point, less is the vapour pressure. Hence, the correct order of vapour pressure wil be
Water < acetone < ether

21. Conjugate base = conjugate acid – H^+
$= H_2 - H^+ = H^-$.

22. The redox changes are
$Mn^{7+} + 5e^- \longrightarrow Mn^{2+}$

$Fe^{2+} \longrightarrow Fe^{3+} + 1e^-$

Equivalent weight of $FeSO_4 = \dfrac{152}{1}$

Milli equivalent of $KMnO_4$ = Meq. Of $FeSO_4$

$\therefore\ 0.05\times 5\times V = \dfrac{2}{152/1}\times 1000$

$\Rightarrow\ V = 52.63$ ml

23. $\dfrac{r}{2}C_2 + \dfrac{r}{2}N_2 = r_{CN}$

$\Rightarrow\ \dfrac{X}{2} + \dfrac{r}{2}N_2 = y\ \Rightarrow\ \dfrac{r}{2}N_2 = y - \dfrac{x}{2}$

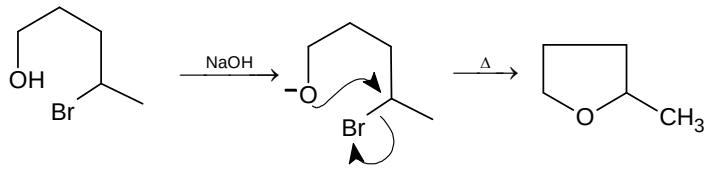
$\Rightarrow \dfrac{r}{2}N_2 = \dfrac{2Y - X}{2}$

$\Rightarrow\ r_{N_2} = 2y - x$

24. Outer elect. Configuration of Co atom = $3d^7 4s^2$
Outer elect. Configuration of Co^{3+} ion = $3d^6$
As F^- is a weak ligand, it cannot cause forcible pairing of electrons within the 3d level. Hence sp^3d^2 hybridization will result outer orbital complex.

25. $\Delta n(g) = n_p - n_r = 5 - 3 = 2$
$\Delta H = \Delta E + \Delta n_g\, RT$
$\Delta E = \Delta H - \Delta n_g\, RT$
$= 19 - (2 \times 2 \times 10^{-3} \times 300)$
$= 19 - 1.2 = 17.8$ K cal.

26. $\Delta G^o = -RT \ln Kp = -(8.314\ J\ k^{-1}\ mol^{-1})\ (900\ K)$
$(2.303)\log(0.05) = 22400\ J\ mol^{-1}$.

27. 

28. 75% of reaction is completed in two half-lives.

29. In the final solution, $[H^+] = (10^{-6} + 10^{-3})/2$
$= 0.5005\times 10^{-3} = 5.005\times 10^{-4}$
$pH = -\log[H^+] = -\log(5.005\times 10^{-4})$
$= 4 - 0.6994 \approx 3.3$

30. $2\,CH_3-\overset{O}{\overset{\|}{C}}-OC_2H_5 \xrightarrow{C_2H_5ONa} H_3C-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-OC_2H_5$
(A)

(A) forms oxime due to keto group, shows tautomerism, also shows idoform test due to $\left(CH_3-\overset{O}{\overset{\|}{C}}-\right)$ group.

Mathematics PART – III

SECTION – A

1. $\frac{1}{2} \le x^2 + \frac{1}{2} < \frac{3}{2}$ $\quad -\frac{1}{2} \le x^2 - \frac{1}{2} < \frac{1}{2}$

$\left[x^2 + \frac{1}{2}\right] = 0,1$ $\quad \left[x^2 - \frac{1}{2}\right] = -1,0$

$\left[x^2 + \frac{1}{2}\right] = 1$ $\quad \left[x^2 - \frac{1}{2}\right] = -1$

$1 \le x^2 + \frac{1}{2} < 2$ $\quad -1 \le x^2 - \frac{1}{2} < 0$

$\frac{1}{2} \le x^2 < \frac{3}{2}$ $\quad -\frac{1}{2} \le x^2 < \frac{1}{2}$

$\frac{1}{\sqrt{2}} \le |x| < \sqrt{\frac{3}{2}}$ $\quad |x| < \frac{1}{\sqrt{2}}$

$$I = \int_{-\sqrt{\frac{3}{2}}}^{-\frac{1}{\sqrt{2}}} \tan^{-1}1\,dx + \int_{\frac{1}{\sqrt{2}}}^{\sqrt{\frac{3}{2}}} \tan^{-1}1\,dx + \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \cot^{-1}(-1)\,dx = \frac{\pi}{4} \times \left(\frac{\sqrt{3}-1}{\sqrt{2}}\right) + \frac{\pi}{4}\frac{\sqrt{3}-1}{\sqrt{2}} + \frac{3\pi}{4} \times \left(\frac{1+1}{\sqrt{2}}\right)$$

$$= \frac{\pi}{4\sqrt{2}}\left[2\sqrt{3} - 2 + 6\right] = \frac{\pi}{2\sqrt{2}}\left(\sqrt{3}+2\right) = \frac{\pi}{2\sqrt{2}}\left(\sqrt{3}+2\right)$$

2. $m \le 2n \le 60 \Rightarrow n \le 2m \le 60$

For m = 1; $1 \le 2n \le 60$ and $n \le 2 \le 60 \Rightarrow \frac{1}{2} \le n \le 2 \Rightarrow n = 1, 2$

For m = 2; $2 \le 2n \le 60$ and $n \le 4 \le 60 \Rightarrow 1 \le n \le 4 \Rightarrow$ n = 1, 2, 3, 4

For m = 3; $3 \le 2n \le 60$ and $n \le 6 \le 60 \Rightarrow \frac{3}{2} \le n \le 6 \Rightarrow$ n = 2, 3, 4, 5, 6

m = 4 n = 2, 3, 4, 5, 6, 7, 8
m = 5 n = 3, 4, …….. 10
m = 6 n = 3, 4 ……12
Total order pair =

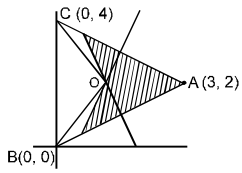
$$\underbrace{2+4}_{m=1\ m=2} + \underbrace{5+7}_{m=3\ m=4} + \underbrace{8+10}_{m=5\ m=6} + \underbrace{11+13}_{m=6\ m=8} + \underbrace{14}_{m=9} + \ldots\ldots + \underbrace{20+22}_{m=13\ m=14} + \underbrace{23+23}_{m=15\ m=16}$$

$$\underbrace{22+22} + \underbrace{21+21} + \underbrace{20+20} + \underbrace{19+19} + \underbrace{18+18} + \underbrace{17+17} + \underbrace{16+16}$$

$$= \underbrace{6+12+18+\ldots\ldots\ldots\ldots+42}_{7\text{ terms}} + 46 + 2(22+21+\ldots\ldots+16)$$

$$= \frac{7}{2} \times (6+42) + 46 + 2 \times \frac{7}{2} \times (22+16) = 7 \times 24 + 46 + 7 \times 38 = 480$$

3. In the figure O is circumcentre and shaded region shows location of P
Distance of P from side BC is maximum if P is at A



4. Let first term is a then $x = a + (m-1)d$, $y = a + (n-1)d$ and $s = \frac{m+n}{2}[2a+(m+n-1)d]$

$(x + y) = 2a + (m + n - 2)d$ and $(x - y) = (m - n)d$

$\Rightarrow d = \left(\frac{x-y}{m-n}\right) \Rightarrow (x + y) = 2a + (m + n - 1)d - \left(\frac{x+y}{m-n}\right)$

$S = \left(\frac{m+n}{2}\right)\left[(x+y)+\frac{x-y}{m-n}\right]$

5. $\sin^{-1}\left\{\frac{1}{2}\left(\frac{1}{x}+x\right)\right\}$ check $f(1^-) = \sin^{-1}1^+$, $f(1) = \sin^{-1}1$, $f(1^+) = \sin^{-1}1^-$

Hence function is discontinuous so non differentiable

6. $\lim\limits_{n\to\infty}\frac{5.5^n+3^n-4^n}{5^n+2^n+27\times 9^n} = 0$, divide numerator and denominator by highest power i.e. (9^n)

7. $P(x_1 + t(x_2 - x_1), y_1 + t(y_2 - y_1))$

$x = \frac{t x_2+(1-t)x_1}{t+1-t} \Rightarrow y = \frac{t y_2+(1-t)y_1}{t+1-t}$

For internal division

$t > 0$ and $(1 - t) > 0 \Rightarrow 0 < t < 1$

8.

x_1	x/r	y_1	y/r
x_2	x	y_2	y
x_3	xr	y_3	yr

Slope of line joining (x_1, y_1) and (x_2, y_2)

$m = \frac{y-y/r}{x-x/r} \Rightarrow$ Slope of line joining (x_2, y_2) and $(x_3, y_3) \Rightarrow m = \frac{yr-y}{xr-x} = \frac{r(y-y/r)}{r(x-x/r)}$

Slope is same hence (x_1, y_1), (x_2, y_2) and (x_3, y_3) are in straight line

9. $\cos A + 2\cos B + \cos C = 2$

$\cos A + \cos C = 2(1 - \cos B)$

$2\cos\frac{A+C}{2}.\cos\frac{A-C}{2} = 2\times 2\sin^2\frac{B}{2}$

$\sin\frac{B}{2}\cos\frac{A-C}{2} = 2\sin^2\frac{B}{2}$, $\cos\left(\frac{A}{2}-\frac{C}{2}\right) = 2\sin\frac{B}{2}$, $\cos\frac{A}{2}\cos\frac{C}{2}+\sin\frac{A}{2}\sin\frac{C}{2} = 2\sin\frac{B}{2}$

$\sqrt{\frac{s(s-a)}{bc}}\sqrt{\frac{s(s-c)}{ab}}+\sqrt{\frac{(s-b)(s-c)}{bc}}\sqrt{\frac{(s-a)(s-b)}{ab}} = 2\sqrt{\frac{(s-a)(s-c)}{ac}}$

$a + c = 2b$ i.e. a, b, c, in AP

10 $f(x) = \int_0^1 \frac{x^2+t^2}{2-t}\,dt = \int_0^1\left(\frac{x^2}{2-t}-t-2+\frac{4}{2-t}\right)dt$

$= \left(-(x^2+4)\ell n|2-t|-\frac{t^2}{2}-2t\right)\Big|_0^1 = (x^2 + 4)\ell n2 - \frac{1}{2} - 2$

i.e. $y = x^2\ \ell n2 + 4\ \ell n2 - \frac{5}{2}$ which is a parabola

11. Since (4, –4) and (9, 6) lie on $y^2 = 4a(x - b)$

$\therefore 4 = a(4 - b)$ and $9 = a(9 - b)$

$\therefore$ a = 1 and b = 0

$\therefore$ The parabola is $y^2 = 4x$

Let the point R be $(t^2, 2t)$

$$\therefore \text{Area of } \Delta PRQ = \frac{1}{2}\begin{vmatrix} 4 & -4 & 1 \\ 9 & 6 & 1 \\ t^2 & 2t & 1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} 4 & -4 & 1 \\ 5 & 10 & 0 \\ t^2-4 & 2t+4 & 0 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} 5 & 10 \\ t^2-4 & 2t+4 \end{vmatrix}$$

$$= \frac{1}{2}(10t + 20 - 10t^2 + 40) = -5t^2 + 5t + 30 = -5(t^2 - t + \frac{1}{4}) + 30 + \frac{5}{4} = -5\left(t - \frac{1}{2}\right)^2 + \frac{125}{4}$$

$\therefore$ Area in largest when $t = \frac{1}{2}$

$\therefore$ Coordinates of the point R are $(\frac{1}{4}, 1)$

12. $$f(x) = \begin{cases} 3 & x = 0 \\ a - x^2 & 0 < x < 1 \\ mx + b & 1 \le x \le 2 \end{cases}$$

For Lagrange's mean value theorem f(x) must be continuous in [0, 2] and differentiable in (0, 2)

Hence, a = 3, m + b = 2

$$f'(x) = \begin{cases} -2x & 0 < x < 1 \\ m & 1 < x \le 2 \end{cases}$$

$\Rightarrow$ m = – 2 and hence b = 4

13. It is rectangular hyperbola $\Rightarrow e = \sqrt{2}$

14. Normal to ellipse is $ax\sec\theta - by\,\text{cosec}\,\theta = a^2 - b^2$

15. $$f'(x) = \cos\left(\frac{x^2 + 2x + 1}{5}\right) = \cos\left(\frac{(x+1)^2}{5}\right)$$

Since $0 \le x \le 2$, $\frac{1}{5} \le \frac{(x+1)^2}{5} \le \frac{9}{5}$

$\cos\frac{(x+1)^2}{5} = 0$ only when $\frac{(x+1)^2}{5} = \frac{\pi}{2}$

$f'(x) > 0$ if $0 < x < \sqrt{\frac{5\pi}{2}} - 1$

$f'(x) < 0$ if $\sqrt{\frac{5\pi}{2}} - 1 < x < 2$

$\therefore x = \sqrt{\frac{5\pi}{2}} - 1$ is a point of local maximum

16. $I = \int \frac{x^5 dx}{\sqrt{1+x^3}}$

$1 + x^3 = t^2$

$3x^2\, dx = 2t\, dt$

$I = \frac{2}{3}\int \frac{(t^2-1)t\, dt}{t} = \frac{2}{3}\int (t^2-1)dt = \frac{2}{3}\left[t^3 \times \frac{1}{3} - t\right] + c$

$= \frac{2}{9}\left[t^3 - 3t\right] + c = \frac{2}{9}t^3 - \frac{2}{3}t + c = \frac{2}{9}(1+x^3)^{3/2} - \frac{2}{3}(1+x^3)^{1/2} + c$

17. $\sum_{r=1}^{n}(r+1)(r+\omega)(r+\omega^2) = \sum_{r=1}^{n}(r^3+1) = \frac{1}{4}n^2(n+1)^2 + n$

18. $z\bar{z} + \alpha\bar{z} + \bar{\alpha}z + \beta = 0$

$z\bar{z} + 2z + 2\bar{z} - 5 = 0$

$z_0 = -2 \qquad r = \sqrt{4+5}$

$r = 3$

$|z+2| = 3$

19. $7\{7(x^2-13)-6\} - \{14 - 2(x^2-13\} + (x^2-13)\{6-(x^2-13)^2\} = 0$

$\alpha + \beta + \gamma + \delta + \phi + 2 = 0$

20. $f(f(x)) = k(x^5 + x)\ (k \neq 0)$

$f'(f(x)) \cdot f'(x) = k(5x^4 + 1) \Rightarrow$ Can be + or –

But it may not be –ve. Because if f(x) $\uparrow$ ing then $f'(f(x)).f'(x) > 0$

and If f(x) $\downarrow$ ing then $f'(f(x)).f'(x) > 0$ i.e. f(x) is either $\uparrow$ ing or $\downarrow$ ing (also k > 0)

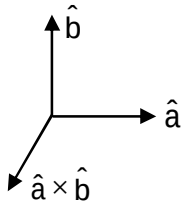
21. Let $\vec{c}$ makes an angle β from $\hat{a} \times \hat{b}$ then $\cos^2\theta + \cos^2\theta + \cos^2\beta = 1$

$2\cos^2\theta = 1 - \cos^2\beta$

$0 \le 1 - \cos^2\beta \le 1$

$0 \le 2\cos^2\theta \le 1 \Rightarrow 0 \le \cos^2\theta \le \frac{1}{2} \Rightarrow -\frac{1}{\sqrt{2}} \le \cos\theta \le \frac{1}{\sqrt{2}}$

$\theta \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$



22. A = (2, –1, 1), B = (1, 2, 3), C = (3, 1, 2)

$\overrightarrow{OA} = 2\hat{i} - \hat{j} + \hat{k}$

$\overrightarrow{OC} = 3\hat{i} + \hat{j} + 2\hat{k}$

$\overrightarrow{OA} \times \overrightarrow{OC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 3 & 1 & 2 \end{vmatrix} = -3\hat{i} - \hat{j} + 5\hat{k}$

$\therefore$ Equation of plane OAC: $3x + y - 5z = d$

Put (0, 0, 0), d = 0

$\therefore$ Plane is $3x + y - 5z = 0$

Shortest distance = $\frac{|3\times 1 + 2 - 5\times 3|}{\sqrt{9+1+25}} = \frac{10}{\sqrt{35}} = 2\sqrt{\frac{5}{7}}$

23. Initially there is one plane
When C_1 is introduced: 1 part is increased
When C_2 is introduced: 2 part is increased
When C_3 is introduced: 4 part is increased
When C_4 is introduced: 8 part is increased

$\therefore$ Total number of parts = 1 + 1 + 2 + 4 + 8 + 16 + ... $= 1 + \frac{1(2^{10}-1)}{2-1} = 1024$

24. for $x \geq 2$, LHS is always non negative and RHS is always –ve
Hence, for $x \geq 2$ no solution
If $1 \leq x < 2$ $(x-2) = (x-1) - 1 = x - 2$ which is an identity
For $0 \leq x < 1$, LHS is '0' and RHS is (–)ve $\Rightarrow$ no solution
$x < 0$, LHS is (+)ve, RHS is (–)ve $\Rightarrow$ no solution

25. Consider $F(x) = \cot\left(\cos^{-1}(|\sin x| + |\cos x|) + \sin^{-1}(-|\cos x| - |\sin x|)\right)$

But $|\sin x| + |\cos x| \in [1, \sqrt{2}]$

$\therefore$ $F(x) = \cot(\cos^{-1}(1) + \sin^{-1}(-1)) = \cot\left(0 - \frac{\pi}{2}\right) = 0 = g(3)$ (As $F(x) = 0, \forall x \in DF$)

26. Consider a fixed face diagonal. It is parallel to corresponding one on the opposite face. It intersects the other diagonal of the same face and one diagonal on each of the four neighbouring faces. The other five face diagonals form five pairs of skew lines with the fixed one. By symmetry, each of the twelve faces diagonals. The total count of pair of skew lines is $12 \times 5 = 60$ but this must be halved since each pair is counted twice.

27. Clearly $N = \phi$
Now M consists of the curve $xy = 1$ and $xy = -1$ in the first and fourth quadrant
Hence, $N \subset M$

28. $y = \frac{1-x}{1+x}$, $F(y) = x = \frac{1-x}{1+x}$

$F(-2-x) = -\left(\frac{3+x}{1+x}\right) = -2 - F(x)$

$F(F(x)) = -x$, $F\left(\frac{1}{x}\right) = F(x)$

29. $p \rightarrow (p \vee q)$ and $q \rightarrow (p \rightarrow q)$ both are tautologies, while formulae
$\sim p \wedge (p \vee q)$ and $\sim p \wedge \sim q$ are not

30. $\bar{x} = 48$, n = 200 standard deviation = 3

Variance = $\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$

$9 = \frac{\sum x^2}{200} - (48)^2$

$\sum x^2 = 46200$